

Quiz 9.6

X is the binomial $(100, 0.5)$ random variable and Y is the discrete uniform $(0, 100)$ random variable. Calculate and graph the PMF of $W = X + Y$.

Further Reading: [Dur94] contains a concise, rigorous presentation and proof of the central limit theorem.

Problems

Difficulty: ● Easy ■ Moderate ♦ Difficult ♦♦ Experts Only

9.1.1 ● X_1 and X_2 are iid random variables with variance $\text{Var}[X]$.

- (a) What is $E[X_1 - X_2]$?
- (b) What is $\text{Var}[X_1 - X_2]$?

9.1.2 ● Flip a biased coin 100 times. On each flip, $P[H] = p$. Let X_i denote the number of heads that occur on flip i . What is $P_{X_{33}}(x)$? Are X_1 and X_2 independent? Define $Y = X_1 + X_2 + \cdots + X_{100}$. Describe Y in words. What is $P_Y(y)$? Find $E[Y]$ and $\text{Var}[Y]$.

9.1.3 ■ A radio program gives concert tickets to the fourth caller with the right answer to a question. Of the people who call, 25% know the answer. Phone calls are independent of one another. The random variable N_r indicates the number of phone calls taken when the r th correct answer arrives. (If the fourth correct answer arrives on the eighth call, then $N_4 = 8$.)

- (a) What is the PMF of N_1 , the number of phone calls needed to obtain the first correct answer?
- (b) What is $E[N_1]$, the expected number of phone calls needed to obtain the first correct answer?
- (c) What is the PMF of N_4 , the number of phone calls needed to obtain the fourth correct answer? Hint: See Example 3.13.
- (d) What is $E[N_4]$? Hint: N_4 can be written as the independent sum $N_4 = K_1 + K_2 + K_3 + K_4$, where each K_i is distributed identically to N_1 .

9.1.4 ■ X_1 , X_2 and X_3 are iid continuous uniform random variables. Random variable $Y = X_1 + X_2 + X_3$ has expected value $E[Y] = 0$ and variance $\sigma_Y^2 = 4$. What is the PDF $f_{X_1}(x)$ of X_1 ?

9.1.5 ■ Random variables X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 2 & x \geq 0, y \geq 0, x+y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

What is the variance of $W = X + Y$?

9.2.1 ● For a constant $a > 0$, a Laplace random variable X has PDF

$$f_X(x) = \frac{a}{2} e^{-a|x|}, \quad -\infty < x < \infty.$$

Calculate the MGF $\phi_X(s)$.

9.2.2 ■ Random variables J and K have the joint probability mass function

$P_{J,K}(j,k)$	$k = -1$	$k = 0$	$k = 1$
$j = -2$	0.42	0.12	0.06
$j = -1$	0.28	0.08	0.04

- (a) What is the MGF of J ?
- (b) What is the MGF of K ?
- (c) Find the PMF of $M = J + K$?
- (d) What is $E[M^4]$?

9.2.3 ■ X is the continuous uniform (a,b) random variable. Find the MGF $\phi_X(s)$. Use the MGF to calculate the first and second moments of X .

9.2.4 ♦ Let X be a Gaussian $(0, \sigma)$ random variable. Use the moment generating function to show that

$$\begin{aligned} E[X] &= 0, & E[X^2] &= \sigma^2, \\ E[X^3] &= 0, & E[X^4] &= 3\sigma^4. \end{aligned}$$

Let Y be a Gaussian (μ, σ) random variable. Use the moments of X to show that

$$\begin{aligned} E[Y^2] &= \sigma^2 + \mu^2, \\ E[Y^3] &= 3\mu\sigma^2 + \mu^3, \\ E[Y^4] &= 3\sigma^4 + 6\mu\sigma^2 + \mu^4. \end{aligned}$$

9.2.5 ♦♦ Random variable K has a discrete uniform $(1, n)$ PMF. Use the MGF $\phi_K(s)$ to find $E[K]$ and $E[K^2]$. Use the first and second moments of K to derive well-known expressions for $\sum_{k=1}^n k$ and $\sum_{k=1}^n k^2$.

9.3.1 ● N is the binomial $(100, 0.4)$ random variable. M is the binomial $(50, 0.4)$ random variable. M and N are independent. What is the PMF of $L = M + N$?

9.3.2 ● Random variable Y has the moment generating function $\phi_Y(s) = 1/(1-s)$. Random variable V has the moment generating function $\phi_V(s) = 1/(1-s)^4$. Y and V are independent. $W = Y + V$.

- What are $E[Y]$, $E[Y^2]$, and $E[Y^3]$?
- What is $E[W^2]$?

9.3.3 ● Let $K_1, K_2, \dots$ denote a sequence of iid Bernoulli (p) random variables. Let $M = K_1 + \dots + K_n$.

- Find the MGF $\phi_K(s)$.
- Find the MGF $\phi_M(s)$.
- Use the MGF $\phi_M(s)$ to find $E[M]$ and $\text{Var}[M]$.

9.3.4 ● Suppose you participate in a chess tournament in which you play n games. Since you are an average player, each game is equally likely to be a win, a loss, or a tie. You collect 2 points for each win, 1 point for each tie, and 0 points for each loss. The outcome of each game is independent of the outcome of every other game. Let X_i be the

number of points you earn for game i and let Y equal the total number of points earned over the n games.

- Find the moment generating functions $\phi_{X_i}(s)$ and $\phi_Y(s)$.
- Find $E[Y]$ and $\text{Var}[Y]$.

9.3.5 ● At time $t = 0$, you begin counting the arrivals of buses at a depot. The number of buses K_i that arrive between time $i - 1$ minutes and time i minutes has the Poisson PMF

$$P_{K_i}(k) = \begin{cases} 2^k e^{-2} / k! & k = 0, 1, 2, \dots, \\ 0 & \text{otherwise.} \end{cases}$$

$K_1, K_2, \dots$ are an iid random sequence. Let $R_i = K_1 + K_2 + \dots + K_i$ denote the number of buses arriving in the first i minutes.

- What is the moment generating function $\phi_{K_i}(s)$?
- Find the MGF $\phi_{R_i}(s)$.
- Find the PMF $P_{R_i}(r)$. Hint: Compare $\phi_{R_i}(s)$ and $\phi_{K_i}(s)$.
- Find $E[R_i]$ and $\text{Var}[R_i]$.

9.3.6 ● Suppose that during the i th day of December, the energy X_i stored by a solar collector is a Gaussian random variable with expected value $32 - i/4$ kW-hr and standard deviation of 10 kW-hr. Assuming the energy stored each day is independent of any other day, what is the PDF of Y , the total energy stored in the 31 days of December?

9.3.7 ■ $K, K_1, K_2, \dots$ are iid random variables. Use the MGF of $M = K_1 + \dots + K_n$ to prove that

- $E[M] = n E[K]$.
- $E[M^2] = n(n-1)(E[K])^2 + n E[K^2]$.

9.4.1 ■ $X_1, X_2, \dots$ is a sequence of iid random variables each with exponential PDF

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

- Find $\phi_X(s)$.

- (b) Let K be a geometric random variable with PMF

$$P_K(k) = \begin{cases} (1-q)q^{k-1} & k = 1, 2, \dots, \\ 0 & \text{otherwise.} \end{cases}$$

Find the MGF and PDF of $V = X_1 + \dots + X_K$.

9.4.2 ■ In any game, the number of passes N that Donovan McNabb will throw is the Poisson (30) random variable. Each pass is completed with probability $q = 2/3$, independent of any other pass or the number of passes thrown. Let K equal the number of completed passes McNabb throws in a game. What are $\phi_K(s)$, $E[K]$, and $\text{Var}[K]$? What is the PMF $P_K(k)$?

9.4.3 ■ Suppose we flip a fair coin repeatedly. Let X_i equal 1 if flip i was heads (H) and 0 otherwise. Let N denote the number of flips needed until H has occurred 100 times. Is N independent of the random sequence $X_1, X_2, \dots$? Define $Y = X_1 + \dots + X_N$. Is Y an ordinary random sum of random variables? What is the PMF of Y ?

9.4.4 ♦ K , the number of passes that Donovan McNabb completes in any game, is the Poisson (20) random variable. If NFL yardage were measured with greater care (as opposed to always being rounded to the nearest yard), officials might discover that each completion results in a yardage gain Y that is the exponential random variable with expected value 15 yards. Let V equal McNabb's total passing yardage in a game. Find $\phi_V(s)$, $E[V]$, $\text{Var}[V]$, and (if possible) the PDF $f_V(v)$.

9.4.5 ♦ This problem continues the lottery of Problem 3.7.10 in which each ticket has 6 randomly marked numbers out of $1, \dots, 46$. A ticket is a winner if the six marked numbers match 6 numbers drawn at random at the end of a week. Suppose that following a week in which the pot carried over was r dollars, the number of tickets sold in that week, K , is the Poisson (r) random variable. What is the PMF of the number of winning

tickets? Hint: What is the probability q that an arbitrary ticket is a winner?

9.4.6 ♦ X is the Gaussian $(1, 1)$ random variable and K is a discrete random variable, independent of X , with PMF

$$P_K(k) = \begin{cases} q(1-q)^k & k = 0, 1, \dots, \\ 0 & \text{otherwise.} \end{cases}$$

Let $X_1, X_2, \dots$ denote a sequence of iid random variables each with the same distribution as X .

- What is the MGF of K ?
- What is the MGF of $R = X_1 + \dots + X_K$? Note that $R = 0$ if $K = 0$.
- Find $E[R]$ and $\text{Var}[R]$.

9.4.7 ♦♦ Let $X_1, \dots, X_n$ denote a sequence of iid Bernoulli (p) random variables and let $K = X_1 + \dots + X_n$. In addition, let M denote a binomial (n, p) random variable, independent of $X_1, \dots, X_n$. Do the random variables $U = X_1 + \dots + X_K$ and $V = X_1 + \dots + X_M$ have the same expected value? Be careful: U is *not* an ordinary random sum of random variables.

9.4.8 ♦♦ Suppose you participate in a chess tournament in which you play until you lose a game. Since you are an average player, each game is equally likely to be a win, a loss, or a tie. You collect 2 points for each win, 1 point for each tie, and 0 points for each loss. The outcome of each game is independent of the outcome of every other game. Let X_i be the number of points you earn for game i and let Y equal the total number of points earned in the tournament.

- Find the moment generating function $\phi_Y(s)$. Hint: What is $E[e^{sX_i} | N = n]$? This is not the usual random sum of random variables problem.
- Find $E[Y]$ and $\text{Var}[Y]$.

9.5.1 ● The waiting time in milliseconds, W , for accessing one record from a computer database is the continuous uniform $(0, 10)$ random variable. The read time R

(for moving the information from the disk to main memory) is 3 milliseconds. The random variable X milliseconds is the total access time (waiting time + read time) to get one block of information from the disk. Before performing a certain task, the computer must access 12 different blocks of information from the disk. (Access times for different blocks are independent of one another.) The total access time for all the information is a random variable A milliseconds.

- What is $E[X]$?
- What is $\text{Var}[X]$?
- What is $E[A]$?
- What is σ_A ? time?
- Use the central limit theorem to estimate $P[A > 116\text{ms}]$.
- Use the central limit theorem to estimate $P[A < 86\text{ms}]$.

9.5.2 ● Internet packets can be classified as video (V) or as generic data (D). Based on a lot of observations taken by the Internet service provider, we have the following probability model: $P[V] = 3/4$, $P[D] = 1/4$. Data packets and video packets occur independently of one another. The random variable K_n is the number of video packets in a collection of n packets.

- What is $E[K_{100}]$, the expected number of video packets in a set of 100 packets?
- What is $\sigma_{K_{100}}$?
- Use the central limit theorem to estimate $P[K_{100} \geq 18]$.
- Use the central limit theorem to estimate $P[16 \leq K_{100} \leq 24]$.

9.5.3 ● The duration of a cellular telephone call is an exponential random variable with expected value 150 seconds. A subscriber has a calling plan that includes 300 minutes per month at a cost of \$30.00 plus \$0.40 for each minute that the total calling time exceeds 300 minutes. In a certain month, the subscriber has 120 cellular calls.

- Use the central limit theorem to estimate the probability that the subscriber's bill is greater than \$36. (Assume that the durations of all phone calls are mutually independent and that the telephone company measures call duration exactly and charges accordingly, without rounding up fractional minutes.)
- Suppose the telephone company does charge a full minute for each fractional minute used. Re-calculate your estimate of the probability that the bill is greater than \$36.

9.5.4 ● Let $K_1, K_2, \dots$ be an iid sequence of Poisson (1) random variables. Let $W_n = K_1 + \dots + K_n$. Use the improved central limit theorem approximation to estimate $P[W_n = n]$. For $n = 4, 25, 64$, compare the approximation to the exact value of $P[W_n = n]$.

9.5.5 ● In any one-minute interval, the number of requests for a popular Web page is a Poisson random variable with expected value 300 requests.

- A Web server has a capacity of C requests per minute. If the number of requests in a one-minute interval is greater than C , the server is overloaded. Use the central limit theorem to estimate the smallest value of C for which the probability of overload is less than 0.05.
- Use MATLAB to calculate the actual probability of overload for the value of C derived from the central limit theorem.
- For the value of C derived from the central limit theorem, what is the probability of overload in a *one-second* interval?
- What is the smallest value of C for which the probability of overload in a one-second interval is less than 0.05?
- Comment on the application of the central limit theorem to estimate the overload probability in a one-second interval and in a one-minute interval.

9.5.6 ● Integrated circuits from a certain factory pass a certain quality test with probability 0.8. The outcomes of all tests are mutually independent.

- What is the expected number of tests necessary to find 500 acceptable circuits?
- Use the central limit theorem to estimate the probability of finding 500 acceptable circuits in a batch of 600 circuits.
- Use MATLAB to calculate the actual probability of finding 500 acceptable circuits in a batch of 600 circuits.
- Use the central limit theorem to calculate the minimum batch size for finding 500 acceptable circuits with probability 0.9 or greater.

9.5.7 ■ Internet packets can be classified as video (V) or as generic data (D). Based on a lot of observations taken by the Internet service provider, we have the following probability model: $P[V] = 0.8$, $P[D] = 0.2$. Data packets and video packets occur independently of one another. The random variable K_n is the number of video packets in a collection of n packets.

- What is $E[K_{48}]$, the expected number of video packets in a set of 48 packets?
- What is $\sigma_{K_{48}}$, the standard deviation of the number of video packets in a set of 48 packets?
- Use the central limit theorem to estimate $P[30 \leq K_{48} \leq 42]$, the probability of between 30 and 42 voice calls in a set of 48 calls.
- Use the De Moivre–Laplace formula to estimate $P[30 \leq K_{48} \leq 42]$.

9.5.8 ■ In the presence of a headwind of normalized intensity W , your speed on your bike is $V = 20 - 10W^3$ mi/hr. The wind intensity W is a continuous uniform $(-1, 1)$ random variable. Moreover, the wind changes every ten minutes. Let W_i denote the headwind intensity in the i th ten-minute interval. In a five-hour bike ride,

with 30 ten-minute intervals, the wind intensities $W_1, \dots, W_{30}$ are independent and identical to W . The distance you travel is

$$X = \frac{V_1 + V_2 + \cdots + V_{30}}{6}.$$

Use the CLT to estimate $P[X \geq 95]$.

9.5.9 ■ An amplifier circuit has power consumption Y that grows nonlinearly with the input signal voltage X . When the input signal is X volts, the instantaneous power consumed by the amplifier is $Y = 20 + 15X^2$ Watts. The input signal X is the continuous uniform $(-1, 1)$ random variable. Sampling the input signal every millisecond over a 100-millisecond interval yields the iid signal samples $X_1, X_2, \dots, X_{100}$. Over the 100 ms interval, you estimate the average power of the amplifier as

$$W = \frac{1}{100} \sum_{i=1}^{100} Y_i$$

where $Y_i = 20 + 15X_i^2$. Use the central limit theorem to estimate $P[W \leq 25.4]$.

9.5.10 ■ In the face of perpetually varying headwinds, cyclists Lance and Ashwin are in a 3000 mile race across America. To maintain a speed of v miles/hour in the presence of a w mi/hr headwind, a cyclist must generate a power output $y = 50 + (v + w - 15)^3$ Watts. During each mile of road, the wind speed W is the continuous uniform $(0, 10)$ random variable independent of the wind speed in any other mile.

- Lance rides at constant velocity $v = 15$ mi/hr mile after mile. Let Y denote Lance's power output over a randomly chosen mile. What is $E[Y]$?
- Ashwin is less powerful but he is able to ride at constant power $\hat{y}$ Watts in the presence of the same variable headwinds. Use the central limit theorem to find $\hat{y}$ such that Ashwin wins the race with probability $1/2$.

9.5.11 ♦ Suppose your grade in a probability course depends on 10 weekly quizzes. Each quiz has ten yes/no questions, each

worth 1 point. The scoring has no partial credit. Your performance is a model of consistency: On each one-point question, you get the right answer with probability p , independent of the outcome on any other question. Thus your score X_i on quiz i is between 0 and 10. Your average score, $X = \sum_{i=1}^{10} X_i/100$ is used to determine your grade. The course grading has simple letter grades without any curve: $A: X \geq 0.9$, $B: 0.8 \leq X < 0.9$, $C: 0.7 \leq X < 0.8$, $D: 0.6 \leq X < 0.7$ and $F: X < 0.6$. As it happens, you are a borderline B/C student with $p = 0.8$.

- What is the PMF of X_i ?
- Use the central limit theorem to estimate the probability $P[A]$ that your grade is an A .
- Suppose now that the course has “attendance quizzes.” If you attend a lecture with an attendance quiz, you get credit for a bonus quiz with a score of 10. If you are present for n bonus quizzes, your modified average

$$X' = \frac{10n + \sum_{i=1}^{10} X_i}{10n + 100}$$

is used to calculate your grade: $A: X' \geq 0.9$, $B: 0.8 \leq X' < 0.9$, and so on. Given you attend n attendance quizzes, use the central limit theorem to estimate $P[A]$.

- Now suppose there are no attendance quizzes and your week 1 quiz is scored an 8. A few hours after the week 1 quiz, you notice that a question was marked incorrectly; your quiz score should have been 9. You appeal to the annoying prof who says “Sorry, all regrade requests must be submitted immediately after receiving your score. But don’t worry, the probability it makes a difference is virtually nil.” Let U denote the event that your letter grade is unchanged because of the scoring error. Find an exact expression for $P[U]$.

9.6.1 ● W_n is the number of ones in 10^n independent transmitted bits, each equiprobably 0 or 1. For $n = 3, 4, \dots$, use the `binomialpmf` function to calculate

$$P[0.499 \leq W_n/10^n \leq 0.501].$$

What is the largest n for which your MATLAB installation can perform the calculation? Can you perform the exact calculation of Example 9.14?

9.6.2 ● Use the MATLAB `plot` function to compare the Erlang (n, λ) PDF to a Gaussian PDF with the same expected value and variance for $\lambda = 1$ and $n = 4, 20, 100$. Why are your results not surprising?

9.6.3 ● Recreate the plots of Figure 9.3. On the same plots, superimpose the PDF of Y_n , a Gaussian random variable with the same expected value and variance. If X_n denotes the binomial (n, p) random variable, explain why for most integers k , $P_{X_n}(k) \approx f_Y(k)$.

9.6.4 ● Find the PMF of $W = X_1 + X_2$ in Example 9.17 using the `conv` function.

9.6.5 ■ Use `uniform12.m` to estimate the probability of a storm surge greater than 7 feet in Example 10.4 based on:

- 1000 samples,
- 10000 samples.

9.6.6 ■ X_1 , X_2 , and X_3 are independent random variables such that X_k has PMF

$$P_{X_k}(x) = \begin{cases} 1/(10k) & x = 1, 2, \dots, 10k, \\ 0 & \text{otherwise.} \end{cases}$$

Find the PMF of $W = X_1 + X_2 + X_3$.

9.6.7 ■ Let X and Y denote independent finite random variables described by the probability and range vectors `px`, `sx` and `py`, `sy`. Write a MATLAB function

`[pw,sw]=sumfinitepmf(px,sx,py,sy)`

such that finite random variable $W = X + Y$ is described by `pw` and `sw`.